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MissionTwin AI

LLM and MARL-Based, Game Theory-Powered
Multi-Agent Mission Planning and Resilience Analysis

PROJECT SPECIFICATIONS REPORT

including Literature Review

TEAM	Eralp Duman (CMPE) · Emre Özüm (CMPE) Taha Berk Yıldız (CMPE) · Emir Ağaoğlu (SENG)
SUPERVISOR	Prof. Dr. Gökçe Nur Yılmaz
JURY MEMBERS	Prof. Dr. Tolga Kurtuluş Çapın Dr. Mert Burkay Çöteli
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List of Abbreviations

Abbreviation	Meaning
ACM	Association for Computing Machinery
AI	Artificial Intelligence
API	Application Programming Interface
C2	Command and Control
CMPE / SENG	Computer Engineering / Software Engineering (department and course codes)
DAG	Directed Acyclic Graph
EVREN	Name of the national AI platform of the Presidency of Defense Industries (SSB); see Section 1.2.1
FR / NFR	Functional Requirement / Non-Functional Requirement
GPU	Graphics Processing Unit
IEEE	Institute of Electrical and Electronics Engineers
ISO / IEC	International Organization for Standardization / International Electrotechnical Commission
JSON	JavaScript Object Notation
KVKK	Kişisel Verilerin Korunması Kanunu (Personal Data Protection Law No. 6698)
LLM	Large Language Model
LOO	Leave-One-Out
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MPE2	Multi-Agent Particle Environments, maintained by the Farama Foundation
NATO	North Atlantic Treaty Organization
OWASP	Open Worldwide Application Security Project
P50 / P95	50th / 95th percentile
PPO	Proximal Policy Optimization
ROS 2	Robot Operating System 2
SRS	Software Requirements Specification
SSB	Presidency of Defense Industries (Savunma Sanayii Başkanlığı)
VMAS	Vectorized Multi-Agent Simulator

1 Introduction

1.1 Description

MissionTwin AI is a defense-oriented research prototype for simulation-based multi-agent mission planning, cooperative coordination, and decision analysis. The system combines a Large Language Model (LLM) for high-level task planning, Multi-Agent Reinforcement Learning (MARL) using Multi-Agent Proximal Policy Optimization (MAPPO) for cooperative execution, and Shapley Value analysis to estimate individual contributions to collective mission outcomes [1], [2], [3], [4].

The main research question is whether Shapley-informed, fault-aware task allocation changes mission success and robustness as mission priorities, available resources, or agent availability change. The controlled comparison evaluates the same LLM–MAPPO pipeline with and without Shapley-derived information against random and distance-based heuristics.

The project proposal [1] sets four objectives: implement the LLM–MAPPO prototype; compare random and distance-based heuristics with LLM-only and Shapley-informed LLM allocation; evaluate task reassignment as priorities, resources, and agent availability change; and quantify contribution and failure impact using exact Shapley values and leave-one-out analysis.

The initial system configuration uses three simulated agents. Candidate research environments include VMAS Navigation, VMAS Dispersion, and MPE2 Simple Spread. Benchmark-specific policy training or adaptation may be required; the use of multiple environments does not imply that a single trained policy transfers directly between them. The prototype uses synthetic missions and does not perform real drone flight control or operational military decision execution.

1.2 Constraints

The following constraints shape the design, evaluation, and delivery of the prototype.

1.2.1 Economic Constraints

The project has no confirmed financial budget, so it will prioritize open-source tools and models, existing hardware, and potential computing resources such as university facilities, free or low-cost cloud GPUs, or EVREN. EVREN is the national artificial-intelligence platform of Türkiye's Presidency of Defense Industries (SSB), launched in September 2026, with GPU infrastructure in Türkiye and a contribution-based credit model open to academics and students [5]; whether it can be used for this project has not been confirmed. Because access to external resources is not guaranteed, the design assumes neither a single permanent GPU nor paid services, and it does not depend on commercial LLM APIs.

1.2.2 Technical and Computational Constraints

Integrating LLM-based planning, MAPPO coordination, and Shapley analysis introduces computational and integration challenges. LLM selection depends on GPU memory, quantization, context length, and planning latency across available local and cloud resources. A 7–14B open-weight model is the initial evaluation target, subject to performance testing. Different benchmarks may require environment-specific observation mappings and policy retraining. Exact Shapley analysis requires evaluating 2^n coalitions (eight for three agents), with sampling-based approximations considered for larger teams.

1.2.3 Benchmark and Experimental Constraints

Standard benchmark runs must be distinguished from custom priority-change and fault-injection variants, and fair comparisons require matched test cases, separate calibration and held-out evaluation data, and compatible metrics.

A further constraint concerns the Shapley analysis itself: agents in standard benchmarks are often identical and exchangeable, in which case their expected Shapley values are equal and carry little information for allocation.

Evaluation scenarios will therefore include heterogeneous variants (for example, different speeds, sensing ranges, or starting positions), and results for homogeneous settings will be reported separately, including near-uniform Shapley values as a possible negative finding.

1.2.4 Time and Development Constraints

The D2 Project Specifications and Literature Review is due 23 October 2026; subsequent milestones follow the official TED University academic calendar and supervisor-approved schedule [6]. Integration and experimental work will be staged to protect mandatory academic deliverables.

1.2.5 Environmental Constraints

The principal environmental impact is computational energy consumption from MARL training, LLM inference, and repeated simulations. Experiment scheduling, checkpoint reuse, and resource monitoring will be used to reduce unnecessary computation, and compute time (for example, GPU-hours) will be recorded where practical as a transparent proxy for energy use. Any carbon estimates must be labeled as estimates unless measured data are available.

1.2.6 Social and Political Constraints

The work concerns defense-oriented but synthetic multi-agent research. Mission-planning methods of this kind have civilian uses (for example, search and rescue, logistics, and disaster response) as well as defense uses. The project excludes classified data and operational command-and-control (C2) deployment, and its results will be described as research findings in simulation, not as operational capability. Future institutional integration could require separate authorizations, security reviews, and regulatory assessment.

1.2.7 Ethical, Health, and Safety Constraints

The prototype will not command physical unmanned platforms; the ethical and professional obligations of the project are elaborated in Section 1.3. The project involves no human-subject experiments, no personal data, and no physical hardware, so it poses no direct physical health or safety risk to users or third parties. The residual risk is indirect misuse of planning outputs, which is limited by simulation-only execution (NFR-05) and by presenting results as research findings.

1.2.8 Manufacturability and Sustainability Constraints

As a software prototype, the project involves no physical manufacturing; manufacturability and sustainability instead concern portability, reproducibility, maintainability, licensing, and integration effort. Versioned dependencies, modular components, and documented simulation interfaces will support continued development without implying real-system compatibility.

1.2.9 Resource Dependencies and Contingencies

Funding, compute capacity, benchmark compatibility, and implementation effort are interdependent. If access or integration fails, the team will protect core controlled comparisons and formal academic submissions, postpone optional capabilities, and explicitly document changes to research coverage. Table 1 summarizes the principal risks and planned mitigations.

Table 1. Principal project risks and planned mitigations.

Risk	Likelihood / impact	Planned mitigation
Compute access or capacity is unavailable	Medium / High	Prioritize local or university resources; reduce model size or evaluation scope if needed and document protocol deviations.
Benchmark integration takes longer than planned	Medium / Medium	Validate the two core VMAS configurations early; defer MPE2 and optional adapters if needed.
LLM plans are invalid or too slow	Medium / Medium	Validate outputs against the schema and mission constraints, record latency, and use the deterministic distance-based fallback.
Implementation effort threatens core comparisons or submissions	Medium / High	Protect the controlled comparisons and academic deliverables; postpone optional capabilities.

1.3 Professional and Ethical Issues

The team follows the ACM Code of Ethics and Professional Conduct [7], the IEEE Code of Ethics [8], and the ACM/IEEE-CS Software Engineering Code of Ethics and Professional Practice [9], as well as Türkiye's Personal Data Protection Law No. 6698 (KVKK) [10]. Because the project is defense-oriented, it also takes into account the NATO Principles of Responsible Use of Artificial Intelligence (lawfulness; responsibility and accountability; explainability and traceability; reliability; governability; and bias mitigation) [11] and, for the LLM component, the OWASP Top 10 for LLM Applications [12]. The team will explicitly separate experimental findings from claims about real-world operational performance.

- **Avoid harm and preserve human accountability** (ACM 1.2 and 2.5; NATO: responsibility and governability). Mission execution stays inside synthetic simulation, and autonomous physical control and operational targeting are excluded (NFR-05). LLM plans are advisory inputs to a validated simulator; plans, validation outcomes, and fallbacks are logged so that a human reviewer can inspect every decision.
- **Address dual use responsibly.** The team limits the work to generic synthetic scenarios, excludes classified or operationally sensitive data, avoids publishing operationally specific tactics, and consults the supervisor before any external release of code, models, or results.
- **Manage LLM-specific risks** (OWASP LLM01 prompt injection, LLM03 supply chain, LLM05 improper output handling, LLM06 excessive agency, and LLM09 misinformation; NATO: reliability). LLM output is treated as untrusted: it is validated against schemas and mission constraints before execution (FR-03, NFR-01); the LLM has no direct control over the simulator or other tools; free-form mission text is treated as untrusted input to the planner; hallucinated or infeasible plans are rejected and counted; and model weights and dependencies are obtained from official sources and pinned by version.
- **Ensure honest reporting** (ACM 1.3; IEEE Code: honest and realistic claims). The team documents failed training runs, uncertainty, negative results, benchmark modifications, evaluation limitations (including how coalition values are measured), and differences between Shapley attribution and direct failure impact.
- **Protect security and privacy** (ACM 1.6, 1.7, and 2.9; KVKK). Synthetic data are used by default, service credentials and experiment records are safeguarded (NFR-04), and any future use of restricted or personal information would first be reviewed against KVKK and with the supervisor.
- **Promote accessible and fair analysis** (ACM 1.4). Visual outputs will be readable without relying only on color, and the team will assess whether scenario design privileges specific starting positions or roles, since this would bias Shapley-based attribution.
- **Respect intellectual property, licenses, and academic integrity** (ACM 1.5). Third-party frameworks, models, and data sources are cited and their licenses reviewed. Some open-weight model licenses restrict military-related uses in their acceptable-use policies; for example, the Llama 4 Acceptable Use Policy lists "[m]ilitary, warfare, nuclear industries or applications, espionage" among prohibited uses [13]. The license of every candidate model will therefore be checked before selection, and permissively licensed models will be preferred. Reproducible experimental evidence will be provided.

2 Literature Review

This section reviews the research on which MissionTwin AI builds and positions the project against it. Sources were identified through keyword searches (for example, “LLM multi-agent reinforcement learning,” “multi-robot task allocation,” “Shapley value credit assignment,” and “agent failure resilience”) of proceedings pages, publisher sites, and arXiv, giving priority to peer-reviewed work from 2020 onward and to foundational earlier papers; preprints are identified as such. The review is organized into five themes and closes with a positioning summary and the resulting research gap. It is an initial review that will be extended in the Analysis Report.

2.1 LLM-Guided Multi-Agent Planning

Large language models have recently been used as high-level planners for robot and agent teams. L2M2 [3] pairs an LLM for high-level, zero-shot planning with MARL for low-level execution; in VMAS the authors report better performance with less than 20% of the training samples required by baselines. Purely LLM-based systems take a different route. SMART-LLM [14] converts a high-level instruction into a multi-robot plan through task decomposition, coalition formation, and task allocation, each guided by few-shot prompts. RoCo [15] lets each robot’s LLM discuss and agree on a strategy, then grounds the resulting sub-task plans with motion planning and environment feedback. CoELA [16] builds cooperative embodied agents from modular LLM-driven components for perception, memory, communication, planning, and execution.

These systems show that LLMs can decompose instructions and allocate sub-tasks, but they rely largely on prompting and environment feedback rather than on explicit validation of every plan against a schema and the mission constraints, and none of the reviewed LLM-based systems uses explicit estimates of each agent’s contribution. MissionTwin AI adopts the L2M2 hierarchy but adds explicit schema and constraint validation with a deterministic fallback, and it compares LLM planning against non-LLM baselines instead of assuming that it is superior.

2.2 MARL Algorithms, Simulators, and Benchmarks

MAPPO, a PPO variant with a centralized value function, was shown to achieve returns and sample efficiency competitive with, and often better than, many off-policy MARL methods on cooperative benchmarks with minimal tuning [2], which motivates its use as the execution layer here. VMAS [17] is a vectorized, PyTorch-based 2D multi-robot simulator that provides cooperative scenarios, including Navigation and Dispersion, and runs many environments in parallel on a GPU. BenchMARL [18] standardizes MARL training and benchmarking on top of TorchRL, and a TorchRL tutorial [4] demonstrates multi-agent PPO on VMAS Navigation. MPE2 [19], the Farama Foundation’s package of the Multi-Agent Particle Environments, provides lightweight cooperative tasks such as Simple Spread and serves as the extension benchmark. Together these resources make a reproducible MAPPO baseline feasible on modest hardware, although policies must be trained per benchmark and raw returns are not comparable across benchmark families.

2.3 Multi-Robot Task Allocation

Allocating tasks to a team of robots has long been studied as an optimization problem. Gerkey and Matarić [20] classify allocation problems by whether robots perform one or several tasks at a time, whether tasks need one or several robots, and whether assignment is instantaneous or time-extended; Korsah et al. [21] extend this taxonomy to tasks with inter-task dependencies and constraints. As formulated here, each task is assigned to one agent and assignments are recomputed when priorities, resources, or availability change — a constrained assignment problem of the kind these taxonomies describe. For such problems, classical optimization and market-based methods are strong baselines; the distance-based strategy in FR-04 is a simple member of this family, and the LLM-based strategies are evaluated against it.

2.4 Shapley-Based Credit and Contribution Analysis

The Shapley value [22] is the unique allocation rule in cooperative game theory that satisfies the efficiency, symmetry, dummy-player, and additivity axioms; it allocates the value of a coalition among its members according to their average marginal contribution over all orderings. In MARL it has mainly been used for credit assignment during training: Shapley Q-value [23] distributes a global reward among agents, SHAQ [24] derives a Shapley-based value factorization for Q-learning, and Shapley Counterfactual Credits [25] approximate Shapley values by Monte Carlo sampling to assign explicit credit to agent coalitions. Shapley analysis has also reached LLM multi-agent systems: HiveMind [26] uses Shapley values to identify underperforming LLM agents and refine their prompts, and its DAG-Shapley algorithm reduces the number of LLM calls by more than 80%; SHARP [27] and Yang et al. [28] (both arXiv preprints) derive training signals for multi-LLM agent systems from Shapley-based credit.

In all of these works the Shapley value serves as a training or optimization signal, and because exact computation needs 2^n coalition evaluations most rely on approximation. With three agents, MissionTwin AI can compute exact values from eight coalition evaluations, and it uses them for post hoc analysis and as planning input rather than as a reward.

2.5 Resilience to Agent Failure

Robustness to agent loss is usually pursued through training. RADAR [29] trains competing teams of varying size so that a cooperative team stays effective when agents change behavior because of updates or failures, and MAGEC [30] (an arXiv preprint) combines a graph neural network with multi-agent PPO to coordinate a multi-robot patrol under agent attrition and disturbed communication in a ROS 2-based simulator. These approaches embed resilience in the learned policy itself. MissionTwin AI instead keeps the MAPPO policy fixed and handles failures at the planning level: failed agents are marked unavailable, unfinished tasks are reassigned among the remaining agents with the selected strategy (FR-05, FR-08), and the resulting degradation is measured and attributed with Shapley and leave-one-out analysis.

2.6 Positioning and Research Gap

Table 2 summarizes how the reviewed work relates to MissionTwin AI. LLM-based planners address high-level planning, Shapley-based MARL methods address credit assignment during training, and resilience methods address agent loss.

Table 2. Positioning of related work relative to MissionTwin AI, based on each paper's stated contributions ("–" means the aspect is not a stated focus).

Work	LLM planning / allocation	MARL policies	Shapley-based contribution	Agent-failure handling
L2M2 [3]	Yes (high-level)	Yes	–	–
SMART-LLM [14], RoCo [15], CoELA [16]	Yes	–	–	–
Shapley Q-value [23], SHAQ [24], Shapley Counterfactual Credits [25]	–	Yes	Yes (training credit)	–
HiveMind [26], SHARP [27], Yang et al. [28]	Partial (LLM agent teams)	–	Yes (credit and optimization signal)	–
RADAR [29], MAGEC [30]	–	Yes	–	Yes (training-time robustness)
MissionTwin AI (this project)	**Yes (allocation and validation)**	**Yes (MAPPO)**	**Yes (exact; planning input)**	**Yes (replanning)**

Within the sources reviewed for this report, we did not find a study that (i) supplies exact Shapley-based agent-contribution estimates to an LLM planner for task allocation and (ii) evaluates the resulting allocations under injected agent failures with MAPPO policies in matched benchmark scenarios. This is an observation about the reviewed literature, not a claim of novelty; the search will be repeated and extended during the analysis phase. The project's

contribution is therefore framed as a controlled empirical comparison rather than a new algorithm: it asks whether contribution information of this kind improves LLM-based task allocation and reallocation, and the answer may legitimately be negative.

3 Requirements

The system specification contains 11 functional requirements (FR) and 5 non-functional requirements (NFR). FRs define system behavior; NFRs define reliability, performance, reproducibility, security, and execution boundaries.

> **Standards note.** This report uses the SRS organization associated with IEEE Std 830-1998 [31], which has been superseded by ISO/IEC/IEEE 29148, and applies requirement-writing principles (necessary, unambiguous, verifiable) consistent with ISO/IEC/IEEE 29148:2018 [32].

3.1 Functional Requirements (FR)

Table 3. Functional requirements.

ID	Functional requirement	Verification
FR-01	The system shall accept user-submitted mission descriptions as free-form plain text through a designated mission-input field, without requiring a predefined command syntax, and shall pass the submitted text to the LLM planning component as planning context.	Input test (free-form text accepted; text present in the planner context)
FR-02	The system shall accept mission scenarios as JSON documents conforming to a predefined, versioned Mission Scenario Schema that specifies mandatory agent identifiers, task identifiers, task priorities, and resource constraints.	Schema test
FR-03	The system shall generate LLM-based task assignments as JSON documents containing a schema version identifier and forward them to the simulation executor only if they conform to the corresponding Task Assignment Schema and satisfy all constraints defined in the active mission scenario.	Planning test (valid JSON; schema conformance; all active mission constraints satisfied)
FR-04	The system shall allow users to select one of four predefined task allocation strategies (random, distance-based, LLM-only, or Shapley-informed LLM) through the simulation configuration and use the selected strategy to generate task assignments for the simulation run. The Shapley-informed LLM strategy shall provide the contribution values from FR-09 to the LLM planner as additional planning context.	Strategy test (selected strategy applied; Shapley values supplied to the planner for Shapley-informed LLM)
FR-05	The system shall reassign unfinished tasks among available agents when task priorities, resource limits, or agent availability change, using the selected allocation strategy and updated constraints.	Scenario test
FR-06	The system shall support training, saving, and evaluating three-agent MAPPO policies conditioned on assigned task targets.	Policy test
FR-07	The system shall execute at least two predefined benchmark configurations through a common simulation interface.	Integration test
FR-08	The system shall simulate configured agent failures at specified simulation steps by marking the affected agents as unavailable, applying zero actions to them from that step onward, and excluding them from subsequent task allocation and execution. By default, a failed agent remains in the environment as a passive body so that observation and action dimensions stay fixed; removal from the environment is an optional per-scenario setting.	Fault test
FR-09	The system shall calculate and report, for each of the three agents, the exact Shapley value over all eight agent coalitions and the separate leave-one-out (LOO) impact, using identical evaluation scenarios.	Mathematical test (eight coalition values; Shapley values sum to full-team value; separate LOO impact)

ID	Functional requirement	Verification
FR-10	The system shall evaluate all supported allocation strategies using identical evaluation scenarios and the same MAPPO policy checkpoint within each benchmark.	Paired experiment test
FR-11	The system shall record, for each simulation episode, task completion rate, constraint violations, resource consumption, planning latency, scenario ID, random seed, MAPPO checkpoint ID, allocation strategy, LLM identifier and decoding parameters (for LLM-based strategies), LLM fallback activations, agent–task assignments, and simulated failure events with agent IDs and simulation steps.	Record test (all required episode metrics, IDs, settings, assignments, fallbacks, and failures present)

Note: A coalition is a subset of the three agents. The value of a coalition is the mean mission score (task completion rate by default) over the calibration scenarios when only the agents in that coalition are active; the empty coalition has value zero. The LOO impact of an agent is the full-team value minus the value of the team without that agent. These definitions, including the mission score, and the failure protocol of FR-08 are working definitions to be finalized in the Analysis Report.

3.2 Non-Functional Requirements (NFR)

Table 4. Non-functional requirements.

ID	Non-functional requirement	Verification
NFR-01	The system shall continue experiment execution after an invalid LLM response by applying a deterministic fallback allocation that satisfies the active mission constraints (distance-based assignment restricted to feasible agent–task pairs) and shall log each fallback activation.	Recovery test
NFR-02	The system shall complete schema and mission-constraint validation within a P95 latency of 1 second for three-agent scenarios containing up to 20 tasks, measured from assignment receipt to validation decision over 1,000 requests on designated test hardware.	Performance test
NFR-03	The system shall support installation using documented dependency versions and experiment re-execution using saved configurations, random seeds, scenario IDs, LLM identifiers and decoding settings, generated plans, and MAPPO checkpoints.	Reproduction test
NFR-04	The system shall load external service credentials (for example, model-hub, experiment-tracking, or remote-compute access tokens) from environment variables or a secret manager and prevent their values from appearing in version-controlled files or experiment logs.	Security check
NFR-05	The system shall execute mission plans exclusively within supported simulation environments, without transmitting control commands to physical unmanned platforms.	Architecture and integration review

Note: Numerical acceptance thresholds are proposed targets, not measured results, and remain subject to supervisor approval.

3.3 Evaluation Plan

Implementation basis. The planned prototype uses Python and PyTorch, an established MAPPO implementation such as TorchRL or BenchMARL [4], [18], a locally runnable open-weight LLM, versioned synthetic scenarios, automated tests, and saved policy checkpoints. The team will prioritize open-source tools and available local or university computing; access to external compute is not assumed.

Planning model. The LLM will be an open-weight, instruction-tuned model in the 7–14-billion-parameter class (for example, from the Qwen or Mistral families), run locally with quantization if needed. The final choice follows a short feasibility test of valid-JSON rate, planning latency, memory footprint, and license terms (Section 1.3). All LLM-based strategies use the same model and decoding settings, every generated plan is stored for replay (NFR-03), and planning calls are repeated to quantify output variability.

Experimental design.

- **Strategies.** Random and distance-based allocation are compared with LLM-only and Shapley-informed LLM planning (FR-04).

- **Matched conditions.** Within each benchmark, the same scenarios, the same LLM and MAPPO checkpoint, and consistent evaluation settings are used (FR-10). The minimum evaluation covers two benchmark configurations, initially VMAS Navigation and VMAS Dispersion; MPE2 Simple Spread is an extension. Custom fault or priority-change variants are labeled separately from standard benchmark runs.
- **Statistical design.** Train with at least five independent seeds per core benchmark and evaluate at least 100 held-out episodes per strategy and seed. Reducing the seed count (for example, to three) would be documented as a protocol deviation.
- **Heterogeneity.** Because identical agents have equal expected Shapley values, scenarios include heterogeneous variants (Section 1.2.3), and homogeneous results are reported separately.
- **Data separation.** Scenarios used to compute Shapley features (calibration scenarios) are kept separate from held-out test scenarios.

Coalition values and failure protocol. Coalition values are measured by running the fixed, team-trained MAPPO checkpoint with all agents outside the coalition disabled from the first step, using the same mechanism as the failure protocol of FR-08 (zero actions, passive bodies). Because the policy was trained with the full team, these values describe each agent's contribution within that policy, not the performance of a smaller team trained from scratch; this limitation will be stated with the results.

Analysis. Report task completion, constraint violations, planning latency (P50 and P95), simulation time, resource use or clearly labeled cost proxies, failure-induced degradation, reassignment effectiveness, Shapley values, and leave-one-out impact. Summarize variation and uncertainty across seeds, for example with bootstrap confidence intervals [33], and include negative results. Do not compare raw rewards directly across different benchmark families; use within-benchmark absolute or relative differences where appropriate. A positive Shapley-related performance gain is a hypothesis, not a project acceptance condition.

Scope. Table 5 separates the baseline from optional extensions and from excluded work.

Table 5. Project scope.

Category	Content
Baseline (in scope)	Three-agent MAPPO; schema and mission-constraint validation with fallback; replanning after priority, resource, or availability changes; the four allocation strategies; exact three-agent Shapley analysis with a separate leave-one-out measure; controlled failure simulation; two benchmark configurations; experiment records and reports.
Optional extensions	MPE2 evaluation; additional fault models; richer statistics; extra adapters; a dashboard.
Out of scope	Physical-platform control; operational C2 integration; certified mission safety; guaranteed cross-environment policy transfer; a new MARL algorithm; guaranteed improvement over baselines.

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